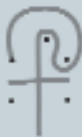




MAKE A MAZE

This simple maze design has been found at ancient sites all over the world, from Finland to Peru. It's very easy to draw one yourself, starting with a cross.

1



Draw a cross and 5 dots as shown here. Make a loop from the top of the cross to the upper left dot.

2



Draw a second loop from the upper right dot to the right side of the cross.

3



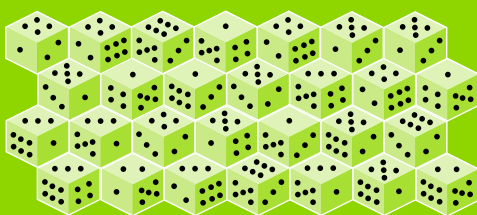
The third loop starts on the left of the cross and ends at the dot below it. Keep this loop wide.

4



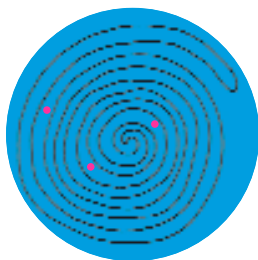
The final loop starts at the lower right dot and ends at the bottom of the cross.

START

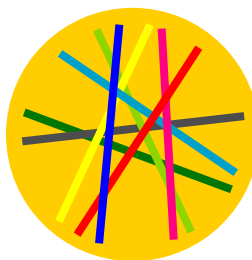


FINISH

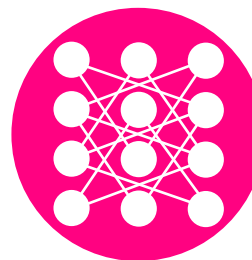
On a normal die, the opposite faces always add up to seven. In the dice maze on the left, some of the dice must be faulty. See if you can find your way across the maze by stepping only on dice that don't add up.



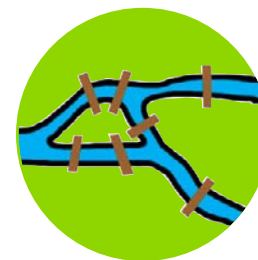
Are the pink dots inside or outside this spiral maze? Can you work out the mathematical rule that tells you whether a dot is inside or outside the maze?



The dark blue stick in the picture above is lying on top of all the other sticks. If you picked off the top sticks one by one, what order would they come off in?



Can you find a route through the maze above that visits every circle once only? Here's a hint: start in the middle of the top row and end in the middle of the bottom row.



The Russian town of Königsberg had seven bridges and two islands. The local people found they could not take a walk that crossed each bridge once only. Leonhard Euler explained why, and that was the start of a branch of maths called "network theory". Try it. Can you explain why it's impossible?

The world's longest hedge maze is at Longleat House in England. It's made of 16,000 English yew trees and has 2.7 km (1.7 miles) of paths. On average, it takes 90 minutes to find the way through.

